Z^*-Numbers, Data Structures, and Thinking in Machine-Mind Architecture

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***Abstract*—A generally intelligent cognitive machine is deliberative, reflective, adaptive, empathetic and rational; continually aiming for the best responses to complex real-world events. This article describes our efforts towards the realization of such a machine – an embodied machine-mind – endowed with abilities of multisensory processing, commonsense reasoning, reflection, consciousness, and empathy. A procedure for contemplation and comprehension by the machine-mind has been formalized. It uses: a) Z^*-numbers for active-thought abstraction, b) multisensory data-structures to encapsulate objective and subjective real-time inputs, and transport these across framework-modules, and c) dynamic action-descriptor formulations for bespoke behavior, perception, and real-time attention modulation. The defined procedure acknowledges the machine-mind's interest in the subject being comprehended, and covers different levels of thinking (instinctive reactions to self-consciousness reasoning). This investigation contributes to the synthesis of 'thinking machines' for man-machine symbiosis.**

***Index Terms*—Artificial general intelligence, computational theory of perceptions, computing with words, embodied systems, man-machine symbiosis, multisensory data-processing.**

I. Introduction

MOST modern robots are designed for objective tasks that are driven by efficiency and precision, like car-manufacturing and delicate surgeries. These do not involve everyday human-communication dynamics. Experiments [1]–[3] show that during interaction with machines, most people expect concern for their feelings, as they would from other human peers. It thus might be practical to design social robots that interact and adapt with us in familiar ways. The synthesis of such machines, however, derives from our understanding of human nature and subtleties of social interactions.

Social robots require [1]–[3] being self-aware ($\sim$ responsive to subtle emotions), forming first-person event-experiences, behaving rationally, and demonstrating consciousness. Such robots are envisioned to be capable of autonomous mind-processes and bespoke responses to situations. Amongst many other possibilities, these systems could act as empathetic care-givers, and aid semantic-plagiarism detection and differential diagnosis. This article documents our efforts towards the synthesis of a framework for such a social machine – a machine-mind [4], and its working principle.

In comparison to existing "specific-purpose" cognitive architectures [5]–[11], the machine-mind aims to model general contemplation. It uses natural language (thoughts, speech, and text) as its primary mode of knowledge processing. Its key features are: a) consideration of instinctive, reflective and self-conscious responses, commonsense reasoning, and creativity as central to the act of thinking, b) inclusion of 'subjective' event-descriptors into existing objective system-operation equations, c) use of involuntary and volitional attention dynamics to handle real-world stimuli, and d) provision for processing multisensory stimuli and memories. Our work is inspired from Turing's 'thinking machine' [12] and Minsky's theories [2], [3] of cognition.

The effective procedure for contemplation and comprehension of real-world events by the machine-mind, formalized in this article, assimilates paradigms designed particularly for this purpose. These include: a) cortex-inspired macro-components of the machine-mind framework [4], b) Z^*-numbers [15] for abstraction of meanings embedded in 'natural language expressions'[1], c) dynamic data-structures [16] that enclose real-world multimodal information and system-visceral affects, and facilitate all forms of inter-modular exchanges, and, d) formulations [17] for bespoke event-perception, behavior and real-time attention arbitration. Interdependence between objective domain knowledge, commonsense, subjective experiences, and system volition forms the foundation of the procedure. It also refers to brain-processes that are often executed beyond our conscious awareness, like the onset of commonsense, activation of memories and social etiquette, and creative contemplation.

The article begins with notes on the aforesaid elements (Section II); description (Section III) and analysis (Section IV) of the defined procedure, and pointers to future scope of work (Section V); and our conclusions (Section VI).

II. Theory

A. The Machine-Mind Framework

The machine-mind [4], [18] is an anthropomorphic, modular, causal, adaptive, embodied cognitive system. It emulates different levels of contemplation [13], [14] towards comprehension

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[1]In this article, the phrase 'natural language expressions' represents multimodal manifestations of the same in terms of thoughts, speech and text.

of real-world events, to forge sustainable symbiotic relationships with other cognitive machines and humans. The architecture comprises the SG (Sensory Gateway), De (Deducer) and M (Manager) macro-modules. These work in sync towards cognition of a region of interest (*R*), where *R* could be external or internal (an idea or memory) to the system. SG captures multisensory stimuli exuding from *R* and transports system responses to it. M provides core-system functionalities like memory management, data-structure population, and resource arbitration. De is dedicated to: a) incremental semantic processing of inputs ($\subseteq$ *R*), knowledge, memories, ideas and concepts, b) initiation of instinctive and deliberative behavior, self-conscious reflection and creativity, c) continual assessment of the system-state, and d) real-time attention-modulation. [Refer Appendix A for details.]

The following sub-sections address problems of thought-abstraction, transportation of these synopses between framework-modules, and factors that define thoughts.

B. Z*-Numbers

Z*-numbers [15], [18] are augmented Z-numbers [19]–[21], designed to summarize the context-sensitive first-person deep-interpretation of a natural language expression (*S*) through:

$$Z^{*}\left(S\right)=\langle C,T,A,B,AG\rangle \tag{1}$$

where,

C = context = F(semantic boundary of R) = a relevant subset of the real-world,

T = *objective* and *subjective* time-frame pertaining to *S*, where,
objective time-frame = system clock-time
[2]*subjective* time-frame = time-span of recollections and prospective visualizations with respect to *S*

A = subject of *S*,

B = 'belief' [19] in *S* given <*C*, *T*, *A*>, and

AG = set of fundamental emotions constituting the ([3]complex/basic) affects embedded in/aroused by *S* for <*C*, *T*, *A*>; each emotion is identified by a (*affect_name*, *strength*, *valence*) triple.

<*B*, *AG*> summarize the system's affective state, are derived from memories in *T*, are typically articulated (implicitly or explicitly) as adjectives or adverbial phrases, and may be represented as fuzzy numbers. <*C*, *T*> granulate Z*-numbers into [4]'Z*-information sets' [15]. <*C*, *B*, *AG*> facilitate interpretation-disambiguation.

The 6-tuple <*X*, *C*, *T*, *A*, *B*, *AG*> defines a Z*-valuation; a generalized constraint on a subject *X*, implying,

$$\text{Probability }(X\text{ is }A)\text{ is }\left(B{*}^{C}AG\right),\text{given}\,\langle C,\ T\rangle \tag{2}$$

Where, the $*^{C}$ operator integrates interdependent roles of *B* and *AG* in decision-making.

An example: For S = It is raining rather heavily!

C = weather-description – either part of a conversation, or an instantaneous observation made to the self

T = {*objective_time* = system_clock-time, *subjective_time* = {recollections (e.g., memories of rain-experiences, …), future visualizations (e.g., transport issues, illness, …)}}

X = {description of the weather, intensity of rain}

A = {raining, rather heavily}

B = emphatic

AG = [(happy, high, +) | (sad, low-moderate, +) | (disgust, moderate-high, -) | (fright, moderate-high, -) | …]

Thus, $(B *^{\mathrm{C}} AG) \sim$ [ecstatic | unhappy | disgust | worried | …]

Therefore, 'one' of the several possible Z*-numbers of S = Z*(*S*) = <weather-description, {*objective_time* = system_clock-time, *subjective_time* = {recent past, immediate future}}, {raining, rather heavily}, emphatic, (happy, high, +)>.

Z*(*S*) is a perceptual abstraction of *S*; deriving incrementally from available multimodal system-inputs, and consequent affects and recollections. Intuitively, Z*-operations would generate new thoughts ($\sim$ mentalese [22]), concepts, ideas, anticipations, counterfactuals, and affects. Z*-numbers are envisaged as machine-mind system-operands.

C. Data-Structures in the Machine-Mind

'Selective attention' [23] helps embodied cognitive frameworks orient their finite resources to an essential subset of the deluge of multisensory information emanating from the real-world. Kernel data-structures of such systems thus require: a) variable-length fields – to encase varying amounts of run-time sensory data; b) hybrid formats – to facilitate simultaneous processing of multiple data-types, structured and unstructured data, and novel real-time information; and c) binding objective and subjective details of stimuli and consequent system-visceral responses into object clusters. The following multisensory constructs [16] theoretically provide for the aforesaid requirements in the machine-mind: [here, *time* = fixation start-time, $\Delta_{time} = fixation\_duration$, R = *region_of_interest*, $' \; '$ = delimiter between values; Appendices B-C illustrate structure-schematics and their activation in the machine-mind framework, respectively.]

1) Afferent Structures: Afferent structures [16] encapsulate sensory inputs for the processor modules of the machine-mind. These are actuated in the following stages [refer Fig. 7 in Appendix B]:

1) Level_1: Encapsulated fingerprints of stimuli-features (e.g., colour, shape, orientation) of all object(s) in the system's 'Ego-sphere' [24] or *R* and consequent visceral emotions, collected through [*time*, Δ_{time}] in the sensory memories of the SG-units.
2) Level_2: Per-object, feature-bound, spatio-temporal clusters of level_1 granules; spawned by M, in sync with the generation of level_1 structures, and processed by De at the end of a [*time*, Δ_{time}] window. These activate object-recognition and associated memories. Spatio-temporal

[2]The relevant aroused memories are translated into their Z*-equivalents, and the 'conscious attention'[16] delegated is proportional to their importance.

[3]http://www.dictionaryofobscuresorrows.com/ lists complex emotions we often experience but find it difficult to express linguistically.

[4]Given constant values of *C*, *T* and *AG* across a Z*-information set, the Z-number [*Z*(*S*) = <*A*, *B*>] reduction of each of its constituent Z*-numbers assists multi-criteria deliberative decision-making [46], [47] and evolutionary strategy analysis [48].

compilations of level_2 granules, across consecutive [*time*, Δ_{time}] timeframes, represent mental-maps of *R* and ego-sphere kinetics respectively, and construe the system's perception of the context. The *epigenome_header* field [16] of a level_2 granule summarizes the machine-mind's critical innate responses to the data therein. This actuates a feedforward [25] ($\sim$ 'fast thinking [26]') run through the framework, consequently setting the premise for comprehension of *R*.

2) Inter-Module Messages: Inter-module messages [16] support information (memories, queries, results, instructions, etc.) exchanges (unicast, multicast or broadcast) across system-components. These exchanges are recorded in the machine-mind Log to support deliberative, reflective and self-conscious behavioral-assessments. [Refer Fig. 8 in Appendix B.]

3) Efferent Structures: Efferent structures [16] transport sensorimotor microcodes of instinctive and deliberated reactions to *R* from De-modules to requisite SG-units. [Refer Fig. 9 in Appendix B.]

D. Factors Defining Thoughts

Thoughts are functions of entangled ideas, experiences, logic, facts, and commonsense. Thinking involves [27]:

1) Information recollection and event-description, based on the context, concepts, goals, ideas and commonsense.
2) Action-planning – deliberation, identifying contradictions and consequences, commonsense reasoning.
3) Formation and perfection of mental representations.
4) Development of a personal style (e.g., optimistic, curiosity-driven, orthodox) of problem solving.
5) Handling novel events – curiosity and intention-optimization, trial and error analysis, processing revelations and biases, etc.
6) Mechanisms (e.g., grades of consciousness) to control cascades of sensory information, activated memories and ideas. [Refer Fig. 12 in Appendix F.]
7) Seamless transition across time and domain, without conscious internal 'announcements' of such intentions.
8) Use of mentalese [22] to stitch abstractions, analogies and perceptions into new ideas and concepts.

Appendices D-F present formulations [17] and procedure-modules that connect the aforesaid points and the Z*-paradigm into operation-codes underlying thinking in the machine-mind framework. The following section presents our formalized procedure and analyses of the same.

III. Proposed Work: An Effective Procedure for Machine-Mind Contemplation and Comprehension

The conceptualized embodied machine-mind envisions application in social, symbiotic environments. This involves processing real-world multimodal cues (speech, text, gestures, expressions, etc.) to ensure efficient communication across agents in the setting. Flowgraphs in Fig. 1 and Fig. 2 summarize our interpretation of contemplation and comprehension of the dynamic world by the machine-mind.

The defined procedure uses natural language as the primary mechanism of knowledge-integration, and "*reaches … ingenious solutions to … problems through cycles of conscious thought, unconscious thought, illumination … and verification.*" [28]. Comprehension begins with the encapsulation of afferent multi-sensory inputs by SG, from a *region_of_interest* (*R*) [a subset of the real-world] through *fixation_duration*, into object-descriptor granules. These trigger object-recognition, ego-sphere representation, context-identification, and arousal of associated thoughts, memories, knowledge and ideas. De converts all active sensory and semantic information into their Z*-equivalents, and incrementally processes them into comprehension-granules and efferent system-responses. All operations consider the dynamic system-state – deriving from comprehension-depth, interests, goals, choices and intermittent results.

The afferent *efferent_header* units trigger 'fast thinking [26]' through activation of the feedforward operation-pathway of the framework. 'Slow thinking [26]' follows in tandem, via iterations of deliberative and self-conscious processes by De. These cycles continually modulate attention to a relevant subset ($\sim$ *R*) [refer Appendix F] of all the information (stimuli, memories, etc.) available at any given time. Novel elements in *R* influence innate inspiration [8], inquisitorial mentalese [22], voluntary attention arbitration [29] across *objects*, and consequent curiosity-driven responses. Greater the novelty, visceral impact or significance of a stimulus, stronger is the attention deployed and consequent information captured [30]. If the machine-mind fails to arrive at conclusive/creative responses, it contemplates between asking its attachment-figures [14], experts or the Internet for solutions. The framework then reflects over its internal process-results and all external inputs to arrive at an 'ideally' rational response. M and De continually monitor the system to prevent combinatorial thought explosions.

The procedure uses an evolutionary conceptual-similarity based method to handle new words and abstract ideas. Here the system forages through its knowledge-base for semantically analogous concept-matches with the novel inputs. If no such matches are found, a descriptor-token – containing all currently available information for the input – is created. All memories and tokens are strategically updated.

The algorithm provides scope for creative and metaphorical contemplation as well. The described procedure does not elucidate mechanisms of 'subjective time [31]', memory-management, Log-administration, machine-mentalese arousal and subsequent translation into Z*-valuations, Z*-operations, conversion of sensorimotor instructions into SG-microcodes, and 'multi-realm [11]' thinking. Machine-mind modules responsible for these operations have been highlighted. [Refer Fig. 13 in Appendix G for a synopsis of the algorithm; Table V in Appendix D and Appendix E list all formulae and procedures referenced in Fig. 1 and Fig. 2.]

IV. Results And Discussions

15 human test-subjects were presented the scenario in Fig. 3, and they were asked to compile their thoughts and reflections

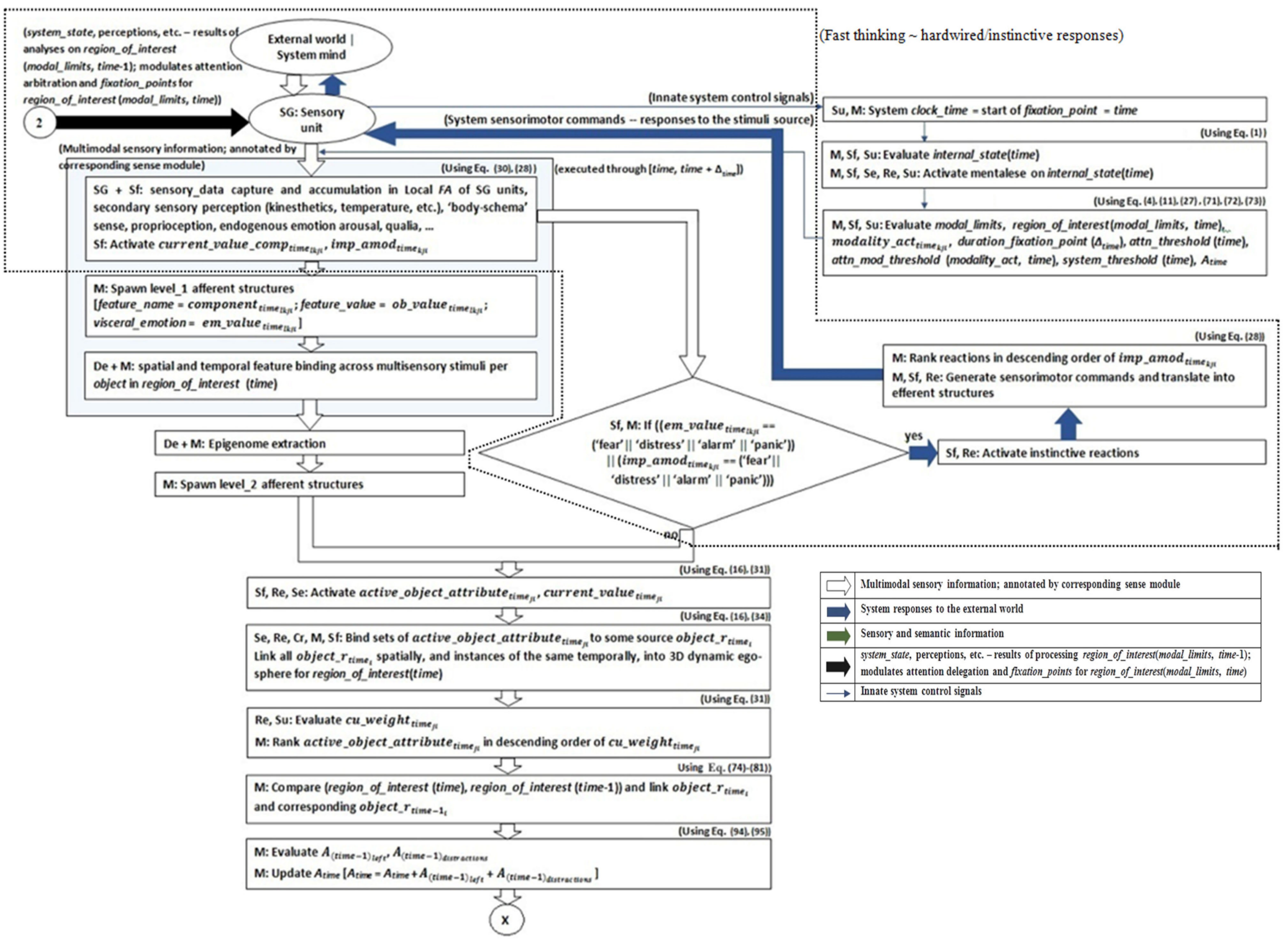

Fig. 1. Flowgraph of Machine-mind operations towards emulation of 'thinking' – Part-I. Here processes pertaining to acquisition of multisensory real-time data and pre-processing of the same towards 'fast thinking'[26] are depicted - instantiation of afferent data-structures, spawning of attention-modulation constructs, and activation of hardwired (intuitive) responses (dotted lines) and their corresponding efferent codes. [Agency codes follow formats in Fig. 6 in Appendix A.]

during its perception. Since none of the responses (verbal-expressions) exhibited atomic steps of contemplation, we annotated them with their inherent affective properties, and used this data to enumerate a subset of the probable response-space for the machine-mind. Words and their usage-frequency, across participants, were recorded [Fig. 4, Fig. 5], as well. Choosing the most-cited response as the desired system behavior, we investigated the nature of inter-modular interactions and operations required for the machine-mind to arrive at the same. Table I and Appendix H describe one out of numerous possible operation-paths and data-structure instantiations towards the required behavior.

[*Note:* Fig. 5 records responses ensuing the 'might dip finger' thought (in Fig. 4). Though 9 individuals used the expression, only 5 acted on it while the others 'waited for someone'. The subsequent thoughts of these 5 subjects were suggestive of multi-level deliberation (ranging from the intuitive 'dip finger to taste' to the reflective "should have 'waited / controlled urge"'; juxtaposition of visceral responses to 'tasty soup' and the self-conscious 'did anyone see'), and were thus chosen as the desired result of machine-mind operations.]

While the previous example demonstrated the self-conscious monologue of the machine-mind, in the following example of differential-diagnosis, the machine-mind plays the role of a medical attendant (D) to a human-patient (P). Here the framework is a social setting where it requires to be an astute observer, and possess senses of proprioception and empathy [32]. One of the numerous possible D-P interaction-scenarios and subsequent framework processes could be:

1) For fixation-point $[t_0, (t_0 + \Delta_0)]$ through saccade $[(t_0 + \Delta_0), (t_0 + \Delta_0 + \partial_0)]$:
   a) Just as P steps into D's chamber, D's afferent structures are instantiated and these instinctively encapsulate multisensory information (e.g, visual and aural details) of P's appearance and D's consequent visceral arousals. Some such data could be "bright red eyes", "flushed face", "persistent cough", and other similar symptoms …
   b) The *epigenome_header* of D's afferent constructs would transfer key multimodal details to De, triggering a stream of fast, intuitive processing. Some ensuing internal thoughts in D could be: "patient looks ill", "wheezing", "rasping cough", "how many days?", "flu?", "red eyes", "high fever?", "rashes?", "appetite?", "headache?", "bronchial congestion?", "Has come before?" …

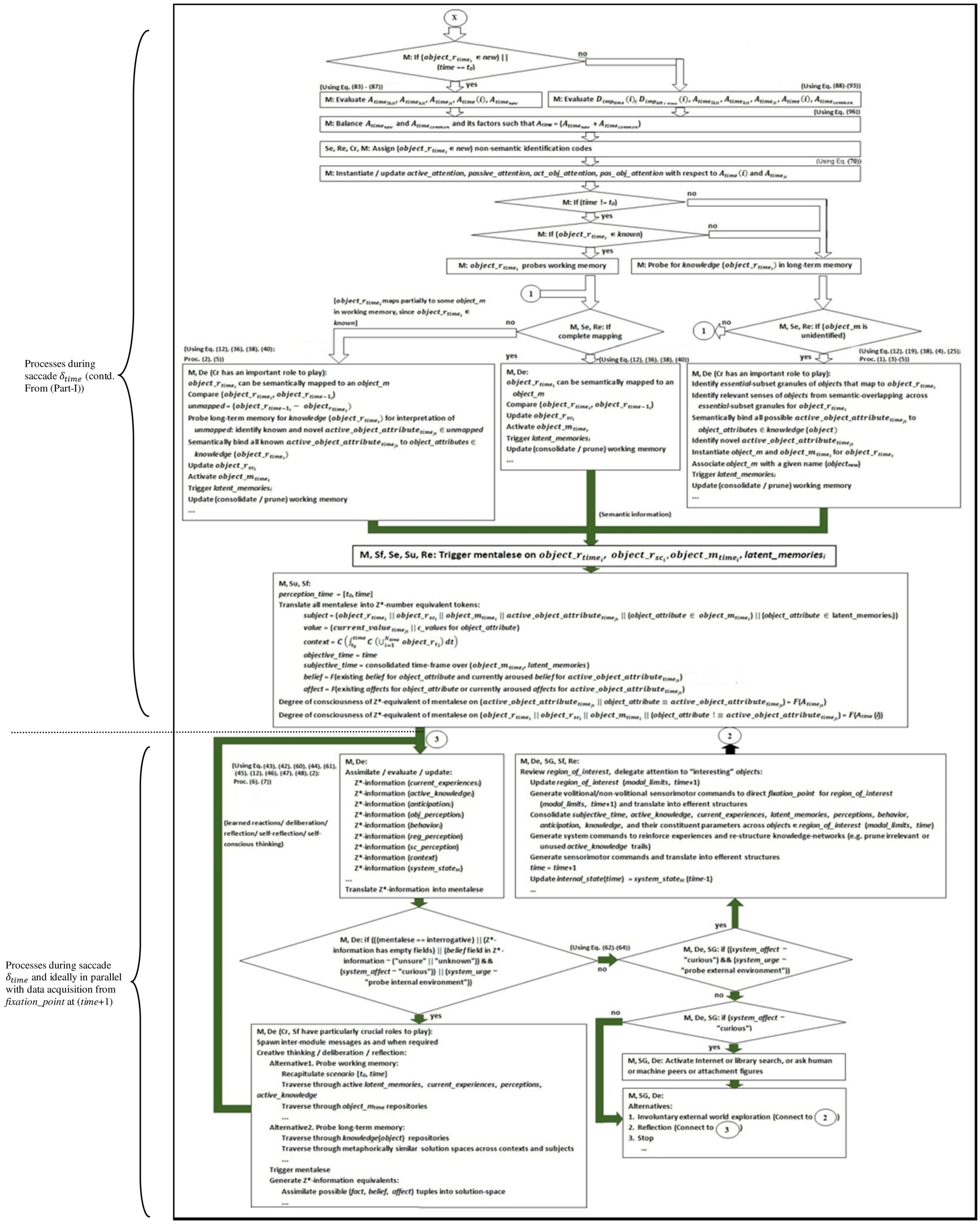

Fig. 2. Flowgraph of Machine-mind operations towards emulation of 'thinking' – Part-II. Here processes on the data acquired, memories and knowledge recollected, towards emulation of 'slow thinking' [26] are depicted - inter-modular exchanges, Z*-number instantiations and procedures thereof, cycles of deliberation and reflection, and triggering of subsequent efferent responses. [Agency codes follow formats in Fig. 6 in Appendix A.]

Fig. 3. The multisensory input – A big blue bowl brimming with a steaming aromatic orange semi-fluid.

c) Operations on the Z^*-expressions of these thoughts should ideally activate *D*'s curiosity on *P*'s condition, whereupon *D* would choose to investigate further.
d) *D*'s efferent response to P could be, "Hello … (possibly shows *P* a chair and *P* sits) You've got a nasty cough! For how long have you been having that condition?" …

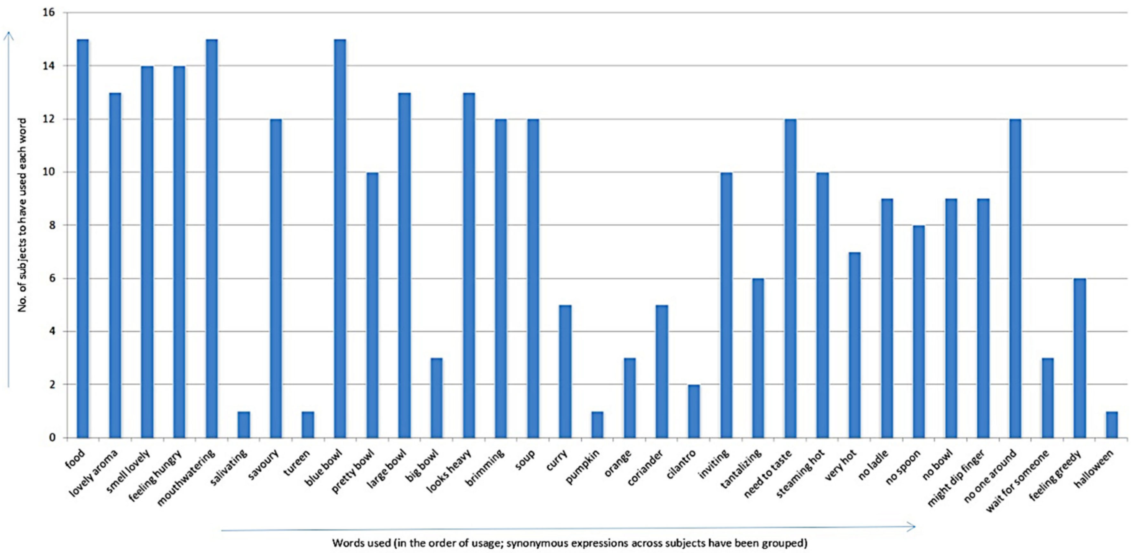

Fig. 4. Plot of words used (X-Axis) versus number of users per word (Y-Axis). Some intuitive observations are: a) natural progression of expressions from visceral responses to the more observatory ones, b) these words are a subset of all possible expressions for the given scenario – it is highly probable that subjects may not have listed all their thoughts, c) evidence of activation of cascades of semantic associations, d) synonymous thoughts and expressions across individuals, with some instances of use of unique vocabulary.

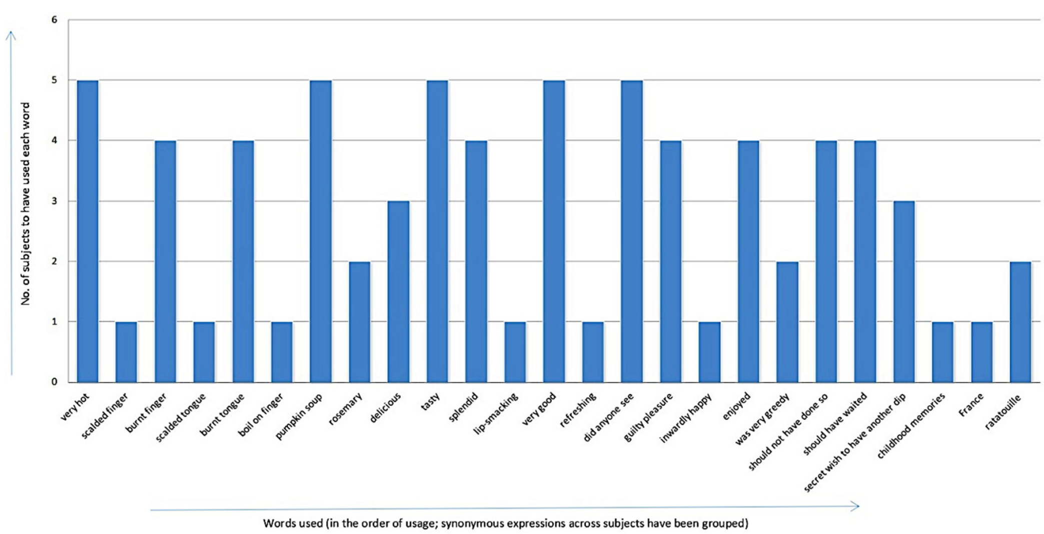

Fig. 5. Count of sub-responses to the decision of 'dip-finger'. Intuitive observations of Fig. 4 hold true here as well.

2) For fixation-point $[t_1, (t_1 + \Delta)]$ through saccade $[(t_1 + \Delta_1), (t_1 + \Delta_1 + \partial_1)]$ and thereafter: Cycles of deliberation and reflection on the information collected (through observations and conversation) and memories activated are executed, until *D* arrives at a confident prognosis or hypothesis of the probable ailment(s). These cycles involve Z^*-conversions of information and operations thereon, information recall, investigations, attention arbitration across stimuli, and intermodular exchanges of semantic and sensory information. If *D* cannot arrive at a solution, it could request for pathological tests and/or refer *P* to a more experienced colleague.

Both the aforedescribed scenarios demonstrate multi-realm deliberation; a glimpse into the depths of our fundamental thoughts and innate capacities as we function in the real-world. Some of these abilities are the encoding of: a) domain knowledge, b) semantic senses of multisensory information (colours, sights, sounds, tastes, smells, textures, temperatures, kinesthetics, proprioception), c) generic concepts of affects and their gestural/linguistic equivalents, and d) social etiquette (e.g., step 1.d of the *D*-*P* interaction example). The machine-mind is assumed to possess all such properties for the execution of the aforesaid experiments. The formal synthesis of these properties is a crucial area of future investigation.

TABLE I
STEPS OF DRY-EXECUTION OF THE DEFINED PROCEDURE OF THINKING AND COMPREHENSION OF A REAL-WORLD EVENT

Time	Processes
Fixation-point: [t_0, ($t_0 + \Delta_0$)]	• Input = $\lvert$*region_of_interest*(*modal_limits*, t_0)$\rvert$ = 5 • *internal_state*(t_0) and its associated terms are deduced; *internal_state* (t_0) = innate system-self
Saccade: [($t_0 + \Delta_0$), ($t_0 + \Delta_0 + \partial_0$)] These steps typically involve iterations of deliberation and reflection.	• Multisensory data from *region_of_interest* and induced affective reactions are assimilated over Δ_0 ○ Level_1 afferent structures are spawned ○ *epigenome_header* is extracted ○ Level_2 afferent structures are instantiated • *active_attributes* and corresponding *current_value* tuples, per object, are created • Attributes are spatio-temporally bound to define five $object\_r_{0_i}$ slots • 'ego-sphere' [24] for region_of_interest (*modal_limits*, t_0) is spawned • *new* = {*object*$_1$, *object*$_2$, *object*$_3$, *object*$_4$, *object*$_5$} • Attention deployment follows (83)-(87) (from Table V. in Appendix D) [intuitively, foreground objects are delegated more attention than background objects] • Each $object\_r_{0_i}$ is assigned a non-semantic identification code <format: *object*$_i$, where i => the i^{th} object> • *active_attention* = {*object*$_1$, *object*$_2$, *object*$_3$}; *passive_attention* = {*object*$_4$, *object*$_5$} • system endeavours mapping $object\_r_{0_i}$ objects to respective *object_m*$_i$ [*object*$_1$ = 'blue bowl', *object*$_2$ = ''yellow semi-solid food component', *object*$_3$ = 'steam', *object*$_4$ = 'table', *object*$_5$ = 'wall'] • $object\_m_{0_i}$ and *latent_memories* in relation to *objects* are fetched into working memory • Mentalese on $object\_r_{0_i}$, $object\_m_{0_i}$, and *latent_memories* are triggered in working memory [e.g., "pretty blue-bowl", "looks like soup", "smells like savoury curry", "steaming!", "big bowl!", "looks heavy", "inviting aroma!", "brimming with food!",...] • Mentalese abstracted into equivalent Z*-numbers [e.g., Z_i = <bowl descriptors, current, {big, blue, heavy, brimming}, definitely, (awe, high, +)>, Z_j = <food descriptors, current, {soup, savoury, hot, aromatic, tasty}, probably, (interested, high, +)>, ...] • The system processes Z*-information sets to arrive at an early perception of *context*, *obj_perception*, *reg_perception*, *sc_perception*, *anticipation* [*context*: input-scene; *obj_perception* = big blue-bowl of hot savoury aromatic semi-solid food components = *sc_perception*] • System assesses *system_state*$_{sc}$(t_0) [*system_state* (t_0) = interested about food, wants to taste, probe scene] • *system_affect* = "curious" and *system_urge* = "probe external environment through touch and taste and enhanced senses of vision and olfaction": ○ *fixation_point*: the bowl of food ○ Duration of *fixation_point* at t_1 (= Δ_1) defined ○ Saccade triggered ○ System attention delegated to *fixation_point* and its boundary ○ Translation of *system_urge* into actions (e.g. dip finger into food and taste) ○ Efferent sensorimotor codes triggered

CONTINUED

Fixation-point: [t_1, ($t_1 + \Delta_1$)]	• Input = $\lvert$*region_of_interest*(*modal_limits*, t_1)$\rvert$ = 3 • *internal_state*(t_1) = *I*(*internal_state*(t_0), *system_state* (t_0))
Saccade: [($t_1 + \Delta_1$), ($t_1 + \Delta_1 + \partial_1$)] These steps typically involve iterations of deliberation and reflection.	• Multisensory data from *region_of_interest* and induced affective system-reactions is collected over Δ_1 ○ Level_1 afferent structures are spawned ○ *epigenome_header* is extracted ○ Level_2 afferent structures are instantiated • *active_attributes* and corresponding *current_value* tuples, per object, are created • Attributes are spatio-temporally bound to define three $object\_r_{1_i}$ slots • ego-sphere for *region_of_interest* (*modal_limits*, t_1) is created • Compare [*region_of_interest* (*modal_limits*, t_1), *region_of_interest* (*modal_limits*, t_0)], link corresponding ($object\_r_{1_i}$, $object\_r_{0_i}$) sets • *left* = {*object*$_4$, *object*$_5$,}; *common* = {*object*$_1$, *object*$_2$, *object*$_3$} • Attention deployment follows (83)-(87) (from Table V. in Appendix D) for *new* and (88)-(96) (from Table V. in Appendix D) for *common, left* and *distractions* • *active_attention* = {*object*$_1$, *object*$_2$, *object*$_3$} • All objects map entirely to working memory contents • In addition to active memories, new $object\_m_{1_i}$ and *latent_memories* associated with the identified objects are fetched into working memory • New mentalese in association with $object\_r_{1_i}$, $object\_m_{1_i}$, and *latent_memories* are triggered in working memory [e.g., "Aah! Roasted-pumpkin soup!", "France-trip!", "very hot!", "should have used spoon", "burnt tongue!", "excellent soup!", "did anyone see?",...] • Z*-number sets for all mentalese abstracted [e.g., Z_i = <food descriptors, {current, past}, {roasted-pumpkin soup, delicious, very hot}, definitely, (excited, high, +)>, Z_j = <action descriptors, {current, past}, {greedy, scalded finger and tongue}, definitely, (guilty, high, –)>,...] • System processes Z*-information sets to arrive at enhanced perception of *context*, *obj_perception*, *reg_perception*, *sc_perception*, *anticipation* [*context*: input-scenario; *obj_perception*: A big blue bowl of steaming delicious roasted-pumpkin soup; *sc_perception*: Was greedy to have dipped finger, should have used spoon, has anyone seen, what would they think?...] • System assesses *system_state*$_{sc}$(t_1) [*system_state* (t_1) = curiosity resolved, guilty of greediness, urge to taste more though, stop]
Outcome: *context*: input-scenario; *obj_perception*: A big blue bowl of steaming delicious roasted pumpkin soup; *sc_perception*: Was greedy to dip finger, should have used spoon, has anyone seen, what would they think?...	

The design of the machine-mind draws from the neuroscience and psychology of endogenous deduction [33], multi-realm thinking [14], attention and perception [34], the 'experiencing self' [26] and 'never-ending' learning [35]. The framework is an anthropomorphic multisensory model of an embodied thinking machine. Contemporary cognitive architectures [5]–[11] are unisensory and can provide support for the processes of learning, deliberation and reflection in the machine-mind. However, the purpose of our investigation being the design of a social machine, emulation of the sense of the self is a necessity [3]. The following points highlight novel features of the machine-mind architecture and the defined algorithm, towards the realization of this property:

1) Consideration of subjective factors of bespoke perception and behavior as central to system-processes (refer Appendix D and E) – an attempt at broadening the paradigm from a solely objective approach to an embodied perspective to cognition-modeling.
2) The Self (Sf) and Creative (Cr) modules in the machine-mind provide scope for emulation of the theory of mind and its derivatives (self-consciousness, qualia [32], proprioception, reflection, empathy, imagination, etc.). This leads to inclusion of the realms of instinctive responses, self-reflection and self-consciousness contemplation, and instantiation of subjective parameters of cognition.
3) The Sensory-Gateway (SG) module allows the framework to have multisensory experiences.
4) The kernel data-constructs of the framework:
    a) Support multimodal data representation – inclusive of their objective and subjective constituents.
    b) Variable-length format facilitates encapsulation of varying amounts of data available at any given *time*.
5) Z^*-valuations substantiate abstraction of information in the memory units of the framework. The paradigm:
    a) Lays the foundation for simulation of the system-self and its spin-offs (qualia, reflection, etc.).
    b) Renders the system dynamic and aids emulation of the 'unified, experiencing [26]' self through the *time* parameter. [Refer (47), (48) in Table V.]
    c) Encases crucial factors of mind-simulation like 'tropism (object-features) and situatedness (context)' [15].
    d) Parametric-instantiations derive from knowledge under categories like 'how', and 'by acquaintance' [15], thereby incorporating socio-cultural elements in its definition.
6) Given a region_of_interest (R) at *time*,
    a) The time complexity of the algorithm is $\propto$ ((((*intricacy* | *novelty*) of R for the system) | (recollection-time for R)) & ((*curiosity* | *interest* | *volition*) of the system in R)).
    b) Intuitively, recollection-time for R is $\propto$ (time-elapsed between ((R, *time*) & system's last encounter with (R | semantically similar contexts))).
    c) The space complexity of the algorithm $\propto$ (((depth & spread) of memory-hypergraphs [4] active at *time*) & (space requirements of intermittent results)).

Intuitively, interpretations and solutions by the machine-mind could be hypothetical, well-rounded, imaginary, radical, orthodox, introspective, or vestiges of unconscious processing. Amongst other factors, the nature of these solutions depends on the system's experiences, creativity and ability for cross-disciplinary concept-pairing.

V. Challenging Issues

This section highlights some crucial challenges towards the realization of general thinking machines, with particular reference to the machine-mind framework and the defined procedure. Solutions to these challenges necessitate algorithmic understanding of processes of contemplation and comprehension in the natural world. These solutions are envisioned to lead to the design of the operating system and foundational architecture for the machine-mind.

Sub-sections *A* and *B* focus on the realization of kernel system processes and substrate framework synthesis respectively; sub-section *C* presents ensuing philosophical questions – the answers to which are envisioned to influence system-design.

A. Major Design Concerns: Kernel Processes in Algorithm

1) Emulation of theory of mind [1], [3] and its derivatives (e.g., sense of unified self [32], affect-arousal, empathy, qualia, sense of time, etc.).
2) Procedures for the run-time extraction of *epigenome_header*, and subsequent activation of the feedforward operation-pathway of the framework. This includes techniques for: a) dynamic modulation of fixation-point durations and saccadic intervals, b) encoding the physics of modal signals (e.g., disparities in sound and light wave-velocities) for binding sensory inputs and instinctive system-responses into level_2 object-descriptor granules, and c) event-simultaneity (e.g., thunder and lightning, clapping) perception.
3) Procedures to encode commonsense, intuitive physics and intuitive psychology [36] of the real-world.
4) Assessment of depth of comprehension at strategic points.
5) Dynamic attention moderation across stimuli; involves emulation of grades of consciousness, interest, and volition.
6) Self-appraisal and evolution of components (memory, procedures, data-structures, etc.) to facilitate adaptation to changing environments, life-long learning and memory augmentation (experience-reinforcement, inclusion of insights, corrections, novel facts, perspectives, etc.).
7) Test of sufficiency of the Z^*-number parameters and the data-structure schematics in terms of their capacities to represent all real-world situations with minimal information loss. Run-time parameter-instantiations and operations thereon are related concerns.
8) Processes to deal with combinatorial thought-explosions, lack of thoughts and reasoning through information-discrepancies, and to strike an effective balance between instinctive and deliberative behavior.
9) Emulation of mindful contemplation and that beyond conscious realization [27] (phenomenon like illumination, unconscious incubation and impasse).
10) Methods to understand and represent homographs; use words and phrases in novel ways, and coin new words.
11) Identification and resolution of biases.

B. Realization of Framework: Some Future Investigations

1) Semaphore-like schemes for resource-arbitration.
2) Non-von Neumann schematics (e.g., dataflow [37] and in-memory [38] systems) for the base architecture.
3) Approximate reasoning [39] approaches for emulation of fast-thinking.
4) Bloom-filter inspired strategies [40] to encapsulate perception-fingerprints of stimuli, in sync with the system's afferent data-structures and knowledge-repositories.
5) Neuromorphic hierarchical temporal memory [41] units to realize and connect the *local* FA systems of the SG units to the Concept-Network (ConN) of the machine-mind.
6) Enumeration of an adaptive set of subjective factors of attachment-figure [14] identification and their importance in decision-making [42]. Intuitively, the context-sensitive quale of these parameters, in addition to other objective factors, should lead to near-rational [42] bespoke choices.
7) Non-linguistic methods for knowledge-consolidation – ideally processing data in their source-sensory formats (e.g., imagining colour-mixing, music-production).
8) The concept of a Machine-IQ [43] or cognitive age for intelligent machines.
9) Reinforcement learning mechanisms that analyze successes and failures, and encode the same in the knowledge base.
10) Processes to initiate exploration across solution spaces towards possible identification of novel solutions and emulation of creativity.

C. Questions of Philosophical Interest

1) What do we precisely mean by 'knowledge' or 'understanding'? When do we realize that we have 'understood'? How do millions of neurons achieve these? Can we think without silent internal speech?
2) What is consciousness and what governs the endogenous arousal of affects and qualia? How different will the qualia of a machine be?
3) What is imagination? Can it be operationalized and fostered in machines?
4) Will the system question, "Who or what am I"? Will it develop a personality of its own and emulate empathy? Would it be inspired by another machine or a human?
5) Do living organisms possess a generic format for multisensory information-representation – supporting the inherent capacity to relate to embedded affects in instrumental music and foreign speech? If so, are our defined afferent formats adequate for the representation of such events?
6) What are the factors that influence self-supervised curiosity? Will the system learn the implications of examining random solutions and being creative?
7) Will a machine ever be able to read between the lines and "arrive at so much from so little" [44]?

VI. Conclusion

This article catalogues our efforts towards the conceptualization of a framework for a social, empathetic "thinking machine" – a machine-mind. An all-inclusive, effective procedure for contemplation and comprehension of real-world events by the machine-mind has been formalized. The framework provides a platform for the extension of existing deliberative models of cognition to that of general contemplation, through the incorporation of the reflective, self-reflective and self-conscious realms of thinking. Towards this, some novel properties/components of the architecture and the defined algorithm are: a) an embodied anthropomorphic multimodal machine-mind framework as the basis of its operations, b) use of natural language expressions for information-consolidation, c) multisensory dynamic data-structures for intra-system exchanges, d) Z^*-numbers as perceptual symbols of natural language expressions, e) the system's sense of self and ensuing subjective factors of perception as central to bespoke comprehension, f) coverage of all levels of contemplation, and g) use of selective attention, in sync with the system's choices, to restrict the deluge of information (sensory, semantic) to an essential subset of the same. The machine-mind is envisioned to support man-machine symbiosis (e.g., sympathetic aid for the elderly, differential diagnosis, semantic plagiarism detection). The defined procedure further encourages constructive thinking on thinking, and the need to explore complex natural processes like real-time spatio-temporal binding of multisensory signals into object-descriptors, qualia and affect arousal, dynamic attention deployment, life-long learning, and unconscious adaptation.

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References

[1] C. Breazeal, *Designing Sociable Robots*. Cambridge, MA, USA: Bradford Book–MIT Press, 2002.
[2] R. Farrell *et al.*, "Symbiotic cognitive computing," *AI Mag.*, vol. 37, no. 3, pp. 81–93, 2016.
[3] J. Wiles, "Will social robots need to be consciously aware? (In the newsletter of the autonomous mental development technical committee)," *IEEE Trans. Auton. Mental Develop.*, vol. 11, no. 2, pp. 14–15, Fall 2014.
[4] R. Banerjee and S. K. Pal, "Text comprehension and the computational mind-agencies," *Natural Comput.*, vol. 14, no. 4, pp. 603–635, 2015.
[5] D. P. Benjamin, D. Lyons, and D. Lonsdale, "ADAPT: A cognitive architecture for robotics," in *Proc. 6th Int. Conf. Cogn. Model.*, 2004, pp. 337–338.
[6] M. Tenorth, A. C. Perzylo, R. Lafrenz, and M. Beetz, "Representation and exchange of knowledge about actions, objects, and environments in the RoboEarth framework," *IEEE Trans. Autom. Sci. Eng.*, vol. 10, no. 3, pp. 643–651, Jul. 2013.
[7] S. Lallee and P. F. M. J. Verschure, "How? Why? What? Where? When? Who? Grounding ontology in the actions of a situated social agent," *Robotics*, vol. 4, pp. 169–193, 2015.
[8] J. Gottlieb, P.-Y. Oudeyer, M. Lopes, and A. Baranes, "Information-seeking, curiosity, and attention: Computational and neural mechanisms," *Trends Cogn. Sci.*, vol. 17, no. 11, pp. 585–593, 2013.

[9] I. Kotseruba and J. K. Tsotsos, "40 years of cognitive architectures: Core cognitive abilities and practical applications," *Artif. Intell. Rev.*, pp. 1–78, 2018, doi: 10.1007/s10462-018-9646-y.

[10] E. Mezghani, E. Exposito, and K. Drira, "A model-driven methodology for the design of autonomic and cognitive IoT-based systems: Application to healthcare," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 1, no. 3, pp. 224–234, Jun. 2017.

[11] P. F. M. J. Verschure, "Distributed adaptive control: A theory of the mind, brain, body nexus," *Biologically Inspired Cogn. Archit.*, vol. 1, pp. 55–72, 2012.

[12] A. M. Turing, "Computing machinery and intelligence," *Mind*, vol. 49, no. 236, pp. 433–460, 1950.

[13] M. Minsky, *The Society of Mind*. New York, NY, USA: Simon and Schuster, 1986.

[14] M. L. Minsky, *The Emotion Machine: Commonsense Thinking, Artificial Intelligence, and the Future of the Human Mind*. New York, NY, USA: Simon and Schuster, 2006.

[15] R. Banerjee and S. K. Pal, "Z*-numbers: Augmented Z-numbers for machine-subjectivity representation," *Inf. Sci.*, vol. 323, pp. 143–178, 2015.

[16] R. Banerjee and S. K. Pal, "Data-structures for multisensory information processing in an embodied machine-mind," *IEEE Trans. Cogn. Develop. Syst.*, vol. 10, no. 3, pp. 726–737, Sep. 2018.

[17] R. Banerjee and S. K. Pal, "A computational model for the endogenous arousal of thoughts through Z*-numbers," *Inf. Sci.*, vol. 405, pp. 227–258, 2017.

[18] R. Banerjee and S. K. Pal, "A machine-mind architecture and Z*-numbers for real-world comprehension," in *Pattern Recognition and Big Data*, A. Pal and S. K. Pal, Eds. Singapore: World Scientific, 2017, pp. 807–844.

[19] L. A. Zadeh, "A note on Z-numbers," *Inf. Sci.*, vol. 181, no. 14, pp. 2923–2932, 2011.

[20] S. K. Pal, R. Banerjee, S. Dutta, and S. S. Sarma, "An insight into the Z-number approach to CWW," *Fundamenta Informaticae*, vol. 124, no. 1/2, pp. 197–229, 2013.

[21] B. Kang, Y. Deng, K. Hewage, and R. Sadiq, "A method of measuring uncertainty for Z-number," *IEEE Trans. Fuzzy Syst.*, vol. 27, no. 4, pp. 731–738, Apr. 2019.

[22] S. Pinker, *The Stuff of Thought: Language as a Window Into Human Nature*. New York, NY, USA: Penguin/Viking, 2007.

[23] E. Yee, E. G. Chrysikou, and S. L. Thompson-Schill, "The cognitive neuroscience of semantic memory," in *Oxford Handbook of Cognitive Neuroscience, Volume 1: Core Topics*. London, U.K.: Oxford Univ. Press, 2013, pp. 353–374.

[24] R. A. Peters, K. A. Hambuchen, and R. E. Bodenheimer, "The sensory ego-sphere: A mediating interface between sensors and cognition," *Auton. Robots*, vol. 26, no. 1, pp. 1–19, 2009.

[25] J. J. Foxe and C. E. Schroeder, "The case for feedforward multisensory convergence during early cortical processing," *Neuroreport*, vol. 16, no. 5, pp. 419–423, 2005.

[26] D. Kahneman, *Thinking, Fast and Slow*. New York, NY, USA: Farrar, Straus and Giroux, 2011.

[27] R. W. Weisberg, *Creativity: Understanding Innovation in Problem Solving, Science, Invention, and the Arts*. Hoboken, NJ, USA: Wiley, 2006.

[28] A. B. Miller, *Einstein, Picasso: Space, Time and the Beauty That Causes Havoc*. New York, NY, USA: Basic Books, 2002.

[29] T. C. Kietzmann and P. König, "Effects of contextual information and stimulus ambiguity on overt visual sampling behavior," *Vision Res.*, vol. 110, no. pt. A, pp. 76–86, 2015.

[30] B. G. Shinn-Cunningham, "Object-based auditory and visual attention," *Trends Cogn. Sci.*, vol. 12, no. 5, pp. 182–186, 2008.

[31] V. Arstila and D. Lloyd, Eds., *Subjective Time: The Philosophy, Psychology and Neuroscience of Temporality*. Cambridge, MA, USA: MIT Press, 2014.

[32] V. S. Ramachandran, *The Tell-Tale Brain: A Neuroscientist's Quest for What Makes Us Human*. New York, NY, USA: Norton, 2010.

[33] W. Schneider and R. M. Shiffrin, "Controlled and automatic human information processing: I. Detection, search and attention," *Psychol. Rev.*, vol. 84, no. 1, pp. 1–66, 1977.

[34] A. M. Treisman and G. Gelade, "A feature-integration theory of attention," *Cogn. Psychol.*, vol. 12, no. 1, pp. 97–136, 1980.

[35] T. Mitchell *et al.*, "Never-ending learning," in *Proc. 29th AAAI Conf. Artif. Intell.*, Austin, TX, USA, Jan. 25–30, 2015, pp. 2302–2310.

[36] B. M. Lake, T. D. Ullman, J. B. Tenenbaum, and S. J. Gershman, "Building machines that learn and think like people," *Behav. Brain Sci.*, vol. 40, 2017, Art. no. e253.

[37] Arvind and D. E. Culler, "Dataflow architectures," *Annu. Rev. Comput. Sci.*, vol. 1, pp. 225–253, 1986.

[38] A. Sebastian *et al.*, "Temporal correlation detection using computational phase-change memory," *Nature Commun.*, vol. 8, no. 1115, pp. 1–10, 2017.

[39] S. K. Pal and D. P. Mandal, "Linguistic recognition system based on approximate reasoning," *Inf. Sci.*, vol. 61, pp. 135–161, 1992.

[40] S. Dasgupta, T. C. Sheehan, C. F. Stevens, and S. Navlakha, "A neural data structure for novelty detection," *Proc. Nat. Acad. Sci. USA*, vol. 115, no. 51, pp. 13093–13098, 2018.

[41] A. M. Zyarah and D. Kudithipudi, "Neuromorphic architecture for the hierarchical temporal memory," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 3, no. 1, pp. 4–14, Feb. 2019.

[42] J. A. Rodger, "QuantumIS: A qualia consciousness awareness and information theory quale approach to reducing strategic decision-making entropy," *Entropy*, vol. 21, no. 2, p. 125, 2019, doi: 10.3390/e21020125.

[43] H.-J. Park, B. K. Kim, and K. Y. Lim, "Measuring the machine intelligence quotient (MIQ) of human-machine cooperative systems," *IEEE Trans. Syst., Man, Cybern. A, Syst. Humans*, vol. 31, no. 2, pp. 89–96, Mar. 2001.

[44] O. Sacks, *Seeing Voices*. New York, NY, USA: Vintage Books, 2000.

[45] B. Mcmurray, "Language at three timescales: The role of real-time processes in language development and evolution," *Topics Cogn. Sci.*, vol. 8, no. 2, pp. 393–407, 2016.

[46] H. Peng, X. Wang, T. Wang, and J. Wang, "Multi-criteria game model based on the pairwise comparisons of strategies with Z-numbers," *Appl. Soft Comput. J.*, vol. 74, pp. 451–465, 2019.

[47] K. Shen, X. Wang, and J. Wang, "Multi-criteria decision-making method based on smallest enclosing circle in incompletely reliable information environment," *Comput. Ind. Eng.*, vol. 130, pp. 1–13, 2019.

[48] B. Kang, G. Chhipi-Shrestha, Y. Deng, K. Hewage, and R. Sadiq, "Stable strategies analysis based on the utility of Z-number in the evolutionary games," *Appl. Math. Comput.*, vol. 324, pp. 202–217, 2018.

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